

Project Overview

In the Kingdom of Oura, you are chosen by the Queen to be sent to the Hogwarts School of Witchcraft and Wizardry. You are granted a scholarship because of your potential. An amount of 20,000 gold will be given to you as a monthly allowance. You live quite far from Hogwarts, so you have to reserve a room in the boarding house which costs 4,000 gold monthly. To make sure that every student at least eats everyday, the school offers a mandatory meal plan subscription in the cafeteria which costs 300 gold per day. To make sure that you don't overspend, you need to keep track of your expenses.

Your task is to create an expense-tracking application. The project will not require a graphical user interface and should make the most out of a text-based user interface -- something we should all be familiar with as we learn about programming in Python.

Start of Budgeting

When the user first uses your program, they should be asked for their name which will be used by the program to make responses that are more centered on the user (i.e. customized feedback). The program then proceeds to day one (1). When the game first starts, the user must start with the following values:

- Day = 1
- Expenses for each category = 0
 - There are 6 categories:
(a) **food and drink**, (b) **transportation**, (c) **healthcare**, (d) **housing and utilities**, (e) **leisure**, and (f) **school**
- Balance = 20,000

Start of the Day

At the start of each day, the following must be performed in order:

1. If it is the start of a new month (i.e. Days 1, 31, 61, 91, 121, ...),
 - a. Add the 20,000 gold to the balance.
 - b. Reset the current month's expenses to 0.
 - c. Deduct 4,000 from the balance (i.e. cost for the boarding house), then add it to the current month's expenses for Housing and Utilities.
2. Deduct 300 from the balance, then add it to the current month's expenses for Food and Drink.
3. The day counter, balance, and the current month's expenses for each category must be displayed.
4. The user is presented a list of actions they can select from, or they can choose to exit the program.

User Actions

Here are all possible actions the user can do:

A. Add expenses

- The expenses can be for the following categories:
(a) **food and drink**, (b) **transportation**, (c) **healthcare**, (d) **housing and utilities**, (e) **leisure**, and (f) **school**
- The user must select a category, then they will be asked for the amount they are about to spend.
- If the amount is lower than or equal to the balance, the current month's expense for the category will be updated. Otherwise, the user will not be able to add the amount, a message must be displayed, and the user will be prompted for the amount again.
- NOTE: The current month's expense for each category resets to 0 every month (i.e. after every 30 days from Day 1).

B. End the day, or month

- The user has the option to end the current day, or month.
- Months are counted with respect to Day 1. A month is counted as 30 days, so the start of a new month is always every after 30 days from Day 1 (i.e. Days 1, 31, 61, 91, 121, ...).
- When ending the day, the player will no longer be able to select any actions for the current day.

If the user selected to add expenses, the program should continue to ask for the next actions after the action is performed. The user must also be asked for another input again if the input is invalid -- for example the number 35 was inputted, which doesn't correspond to anything. An error prompt should be shown to the user assuming there's an error.

Also, please note that the player does not have to add expenses always and may directly choose to end the day or month.

End of the Day

After choosing to end the day or month, the following must be performed **in order**:

1. If the user chose to **end the month**,
 - For every day that has passed, deduct 300 gold from the balance for the meal plan subscription, then add it to the current month's expenses for Food and Drink (e.g. If the current day is 18, the end of the current month will be 30. 12 days would have passed, so your program must automatically deduct 12 days * 300 gold = 3,600 gold from the balance, then add it to the current month's expenses for Food and Drink)
 - Display how many days have passed (e.g. If the current day is 18, the end of the current month will be 30. Thus, 12 days would pass).
 - Update the day counter to the end of the month. (e.g. If the current day is 18, the day counter will be 30.)
2. If the user runs out of money (i.e. balance < 300 if not yet end of month; balance < 0 if end of month), a message will be displayed then the program will end. Steps 3-6 will not be performed anymore.
3. The day counter, and balance must be displayed.
4. If it is the **end of the month**,
 - An expenses report will be displayed. The report must show the expenses for each category for the month, and the percentages of these expenses with respect to the starting balance for the current month.
5. Increment the day counter by 1.
6. The next day will start.

End of Program

The program will end

- If the user selects "Exit". Do not forget to implement an exit function.
- OR, if the user runs out of money before the end of the month.

At the end of the program, display the last day and show a report of the total money earned/received, total expenses, and the percentages of the total expenses and remaining balance with respect to the total money earned/received.

Sample Screen Output

```
***** Expenses Tracker *****
Welcome to the Oura Kingdom! I see that you are new here. What is your name?
Name: Pau

    Hello, Pau! You are chosen by Queen Hermione to be sent to Hogwarts
    School of Witchcraft and Wizardry, granting you a scholarship due to your
    potential. An amount of 20,000 gold will be given to you as a monthly allowance.
    Since you live quite far from Hogwarts, we have reserved a room for you in the
    boarding house inside the campus which costs 4,000 gold monthly. To make sure
    that you at least eat everyday, we have arranged the meal plan subscription
    in the cafeteria for you which costs 300 gold per day. If you compute the
    total default expenses which are housing, and food and drink, you still have
    some allowance for other things like transportation, healthcare, leisure, and
    school. But remember that you only have exactly 20,000 gold for all your
    expenses monthly, so you have to make sure that you don't overspend.
    Let me help you keep track of your expenses.
*****

*****
Day 1
Balance: 15700
Food and Drink: 300
Transportation: 0
Healthcare: 0
Housing and Utilities: 4000
Leisure: 0
School: 0

What would you like to do today?
1 - add Food and Drink expenses
2 - add Transportation expenses
3 - add Healthcare expenses
4 - add Housing and Utilities expenses
5 - add Leisure expenses
6 - add School expenses
7 - end the Day
8 - end the Month
0 - exit the program
Enter the number of your choice: 1
Enter the amount: 1000
Expenses successfully added.

What would you like to do today?
1 - add Food and Drink expenses
2 - add Transportation expenses
3 - add Healthcare expenses
4 - add Housing and Utilities expenses
5 - add Leisure expenses
6 - add School expenses
7 - end the Day
8 - end the Month
0 - exit the program
Enter the number of your choice: 2
Enter the amount: 100
```

Expenses successfully added.

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 3

Enter the amount: 200

Expenses successfully added.

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 4

Enter the amount: 300

Expenses successfully added.

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 5

Enter the amount: 400

Expenses successfully added.

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 6

Enter the amount: 500

Expenses successfully added.

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses

- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 7

At the end of Day 1, you have 13200 gold left.

Day 2

Balance: 12900
Food and Drink: 1600
Transportation: 100
Healthcare: 200
Housing and Utilities: 4300
Leisure: 400
School: 500

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 8

... 28 days have passed ...

At the end of Day 30, you have 4500 gold left.

Food and Drink (50.0%): 10000
Transportation (0.5%): 100
Healthcare (1.0%): 200
Housing and Utilities (21.5%): 4300
Leisure (2.0%): 400
School (2.5%): 500

Day 31

Balance: 20200
Food and Drink: 300
Transportation: 0
Healthcare: 0
Housing and Utilities: 4000
Leisure: 0
School: 0

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 5

Enter the amount: 15000

Expenses successfully added.

What would you like to do today?

- 1 - add Food and Drink expenses
- 2 - add Transportation expenses
- 3 - add Healthcare expenses
- 4 - add Housing and Utilities expenses
- 5 - add Leisure expenses
- 6 - add School expenses
- 7 - end the Day
- 8 - end the Month
- 0 - exit the program

Enter the number of your choice: 8

... 29 days have passed ...

Oh no!

You ate the meals without paying.

Hogwarts is no place for thieves.

The Queen is greatly disappointed.

The scholarship is now cancelled.

On day 60, you quit Hogwarts. During your entire stay, Queen Hermione has bestowed you a total of 40000. You spent a total of 43500 which is 108.7% of the total money you received. You are left with 0 gold.